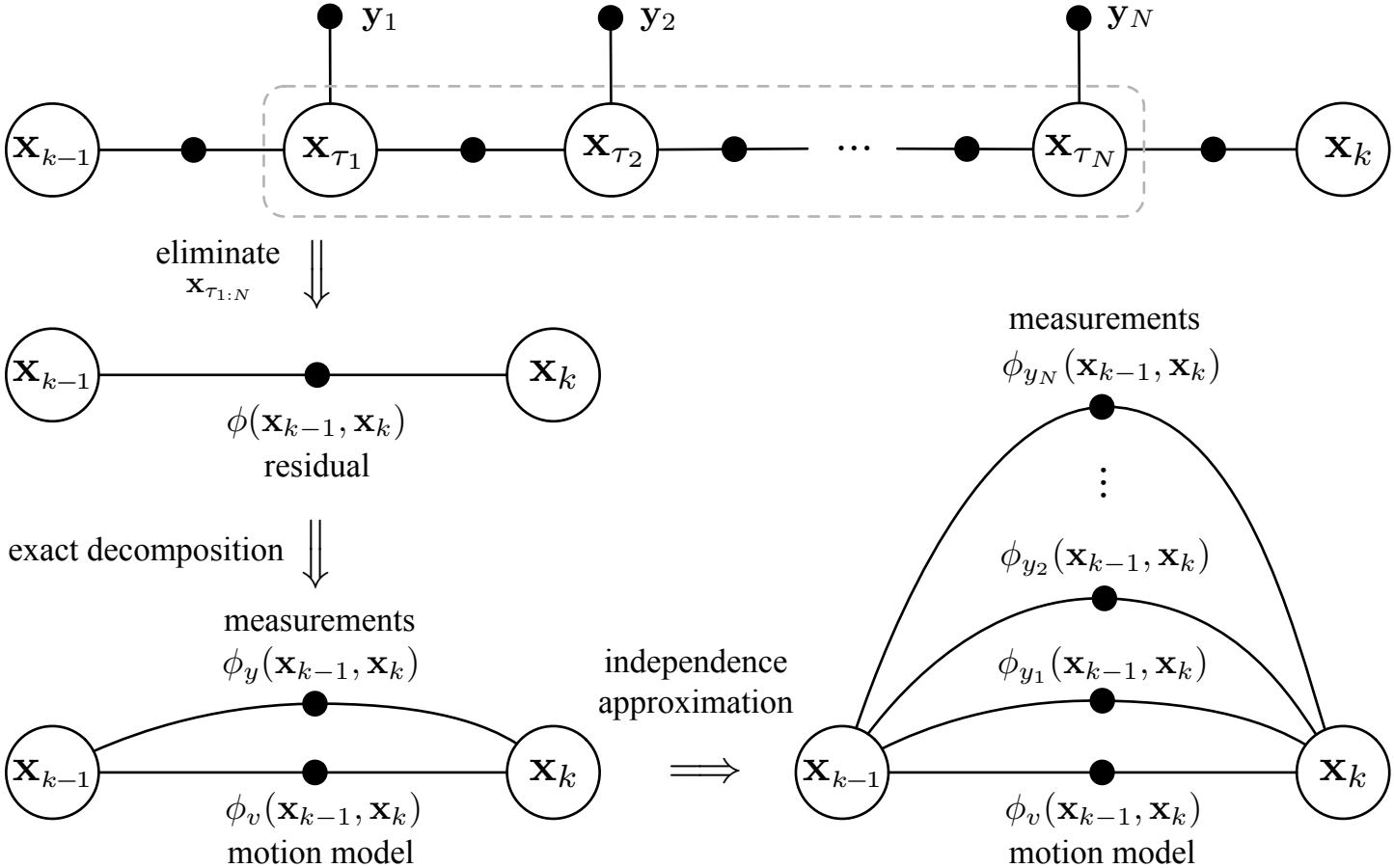
$\mathbf{y}_1$
$\mathbf{y}_2$
$\mathbf{y}_N$
$\mathbf{x}_{k-1}$
$\mathbf{x}_{\tau_1}$
$\mathbf{x}_{\tau_2}$
$\cdots$
$\mathbf{x}_{\tau_N}$
$\mathbf{x}_k$
eliminate
$\mathbf{x}_{\tau_{1:N}}$
$\mathbf{x}_{k-1}$
$\mathbf{x}_k$
$\phi(\mathbf{x}_{k-1}, \mathbf{x}_k)$
residual
exact decomposition
measurements
$\phi_y(\mathbf{x}_{k-1}, \mathbf{x}_k)$
$\mathbf{x}_{k-1}$
$\mathbf{x}_k$
$\phi_v(\mathbf{x}_{k-1}, \mathbf{x}_k)$
motion model
independence
approximation
measurements
$\phi_{y_N}(\mathbf{x}_{k-1}, \mathbf{x}_k)$
$\vdots$
$\phi_{y_2}(\mathbf{x}_{k-1}, \mathbf{x}_k)$
$\phi_{y_1}(\mathbf{x}_{k-1}, \mathbf{x}_k)$
$\mathbf{x}_{k-1}$
$\mathbf{x}_k$
$\phi_v(\mathbf{x}_{k-1}, \mathbf{x}_k)$
motion model